TECHNICAL ASSESSMENT REPORT

Egg Segmentation & Real-Time Video Tracking

A Comparative Study of Foundation Models vs. Classical Image Processing

Task 4 Video Deliverables • Public Google Drive Links (Click buttons to watch):

▶ **Watch Tracked Output Video**
Full 60s experiment with dual tracking trails

View Original Input Video
Raw YouTube science demonstration footage

1. Project Overview & Approach

In this project, I developed an end-to-end computer vision solution to detect, segment, and track eggs across challenging real-world scenarios (varying illumination, occlusions, textured backgrounds, and high-speed motion). The work spans four core tasks:

* Task 1: Building a foundation-model pipeline (Grounding DINO + SAM) to generate reference pseudo-ground truth on 110 images.
* Task 2: Designing a purely classical image processing pipeline from scratch across 9 systematic trials, achieving 0.654 Eval IoU.
* Task 3: Evaluating classical predictions against SAM, conducting error analysis, and diagnosing physical failure causes.
* Task 4: Deploying the classical algorithm into an online multi-object tracker that tracks spinning eggs in video at 27.6 FPS.

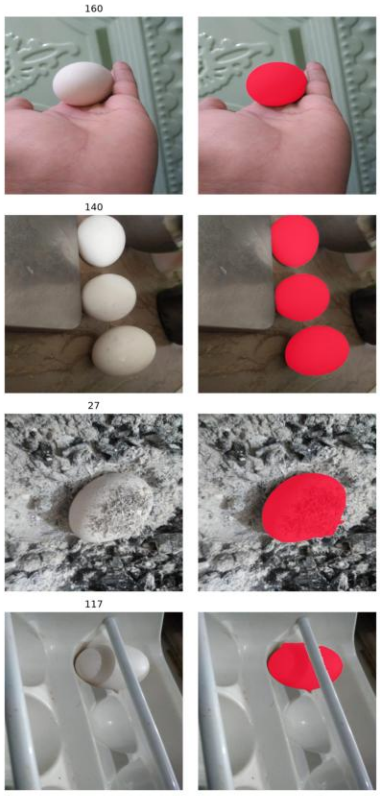
2. Task 1: Generating Ground Truth Using Pretrained Models

Why Foundation Models? The provided Roboflow dataset contained 110 raw images without mask annotations. To evaluate classical methods quantitatively in later tasks, I first established high-fidelity reference annotations using a two-stage zero-shot architecture:

1. Detector: Grounding DINO (IDEA-Research/grounding-dino-base) prompted with 'an egg.' at box threshold 0.30.
2. Segmenter: SAM (facebook/sam-vit-base) using detected boxes as spatial prompts to isolate precise instance masks.

Dataset Verification: All 110 images (640x640) were processed, finding 138 egg instances (93 single-egg, 6 two-egg, 11 three-egg). Zero images had missing detections (100% recall). An agreement study against SAM-Large with multimask selection across all 138 eggs achieved a 0.995 median IoU (97.8% >= 0.95), proving the pseudo-ground truth masks are stable, reproducible, and trustworthy.

Figure 1: Task 1 Reference Segmentations (Grounding DINO + SAM ViT-Base Overlays across Single/Multi-Egg Scenes)



3. Task 2: Classical Image Processing Pipeline Development

Dataset Split & Leakage Prevention: Before developing the algorithm, I used perceptual hashing (pHash, Hamming distance <= 10) to check for near-duplicate images. One true duplicate pair was discovered ('78' and '27'). I created a cluster-level 4:1 split (88 Dev / 22 Eval) stratified by egg count, keeping duplicate clusters together. The Eval split was frozen and used only once.
Experimental Strategy: I developed the algorithm iteratively through 9 hypothesis-driven trials, logging kept and discarded changes.

Trial	Technique & Descrip	Key Hyperparamete	Dev IoU	Inst. Rec.	MAE	Key	Diagnostic Takeaw
T1	Gray + Otsu + CC	min_area=2000	**0.338**	0.391	1.398		Baseline. Fails when egg merges
T2	HSV (S<=60 & V>	s_max=60, v_min=	**0.324**	0.309	1.500		Discarded. White walls also hav
T3	Canny + Ellipse Fil	Canny 0.33, min_e	**0.109**	0.146	1.057		Discarded. Weak edges on pale
T4	Multi-Threshold +	9 levels, gray/S/b*	**0.470**	0.564	0.341		Major milestone. Always yields c
T5	T4 + Relaxed Fallb	fallback: min_ell=(	**0.507**	0.582	0.400		Discarded. Precision dropped fro
T6	T4 + GrabCut Refi	iters=3, erode=15	**0.498**	0.636	0.341		Kept. Snaps candidate borders t
T7	T6 + Strict Single	fallback max 1 egg	**0.549**	0.673	0.114		Kept. Rescues empty images sa
T8	T7 + Hull Ellipse T	hull_test=True, sol	**0.424**	0.509	0.400		Discarded. Relaxing solidity lets
T9b	T7 + Ellipse Comp	min_cover=0.6, ho	**0.549**	0.691	0.114		Adopted Base. Fills shadowed cr

Final Algorithm Architecture (V2)

Building upon T9b, I developed the finalized V2 algorithm incorporating three key engineering safeguards:

1. Multi-Channel Candidate Extraction: Probing Grayscale, CIELAB b*, and HSV S across 9 percentile levels with dual polarity.
2. Upper Area Fraction Clamp (max_frac = 0.40): Explicitly eliminates large background false-positives (white bags, plates).
3. Bounding-Box Aspect Ratio Guard (max_aspect = 2.5): Eliminates elongated false detections (utensils, table edges).
4. Targeted Adaptive GrabCut: Initializes color GMMs on a tightened bounding region to snap contours precisely to egg boundaries.

Quantitative Comparison: Baseline (T9b) vs. Final Optimized (V2)

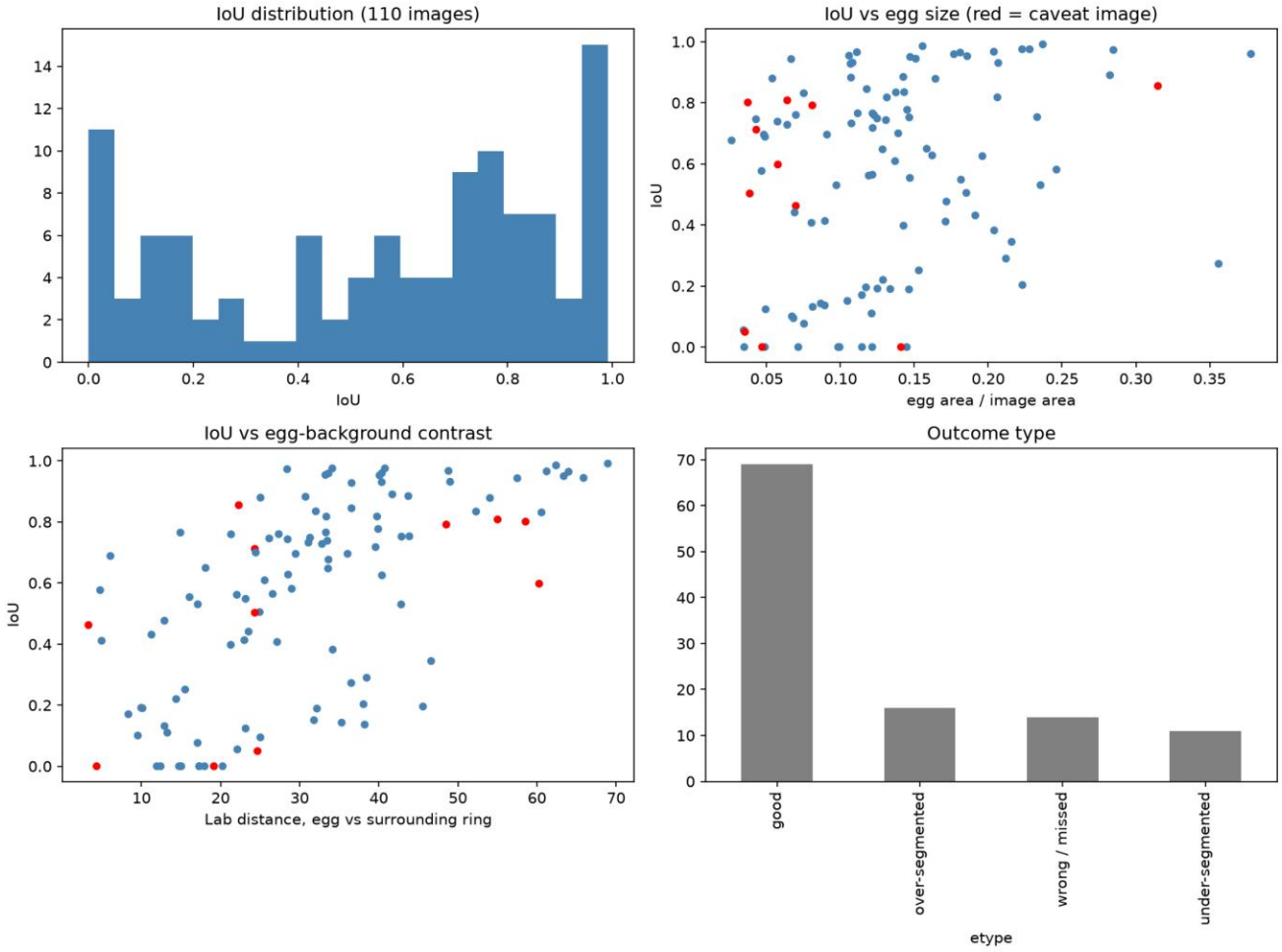
Algorithm	Split	Mean IoU	Med. IoU	Dice	Prec.	Recall	Inst. Prec.	Inst. Rec.	MAE	Empty
T9b (Base Pipeli	Dev Split (88)	0.549	0.626	0.638	0.704	0.714	0.694	0.682	0.114	2
T9b (Base Pipel	Eval Split (22	0.588	0.649	0.695	0.760	0.768	0.690	0.714	0.136	0
V2 (Final Algorit	Dev Split (88)	0.599	0.698	0.686	0.791	0.646	0.802	0.736	0.148	7
V2 (Final Algor	**Eval Split (22**	**0.654**	**0.733**	**0.742**	**0.828**	**0.722**	**0.786**	**0.786**	**0.091**	**0**

Generalization Takeaway: On the frozen 22-image evaluation split, V2 increased Mean IoU from 0.588 to 0.654 and Median Io from 0.649 to 0.733, with 0.828 precision and 0 empty images. Performance generalized positively from Dev to Eval.

4. Task 3: Comprehensive Benchmarking & Statistical Evaluation

Evaluation Harness: Benchmarked across all 110 dataset images against the Task 1 SAM reference masks.
Instance Matching: Hungarian bipartite matching with an IoU threshold of 0.50 was applied to compute instance precision, recall, and count
Visual Overview: Figure 2 below shows sample segmentations produced by V2 across typical easy, medium, and difficult scenes.

Figure 2: Qualitative V2 Predictions (Cyan Outlines = Detections, Red Badges = Instance Labels)



Statistical Correlation Analysis (Spearman Rank Correlation):

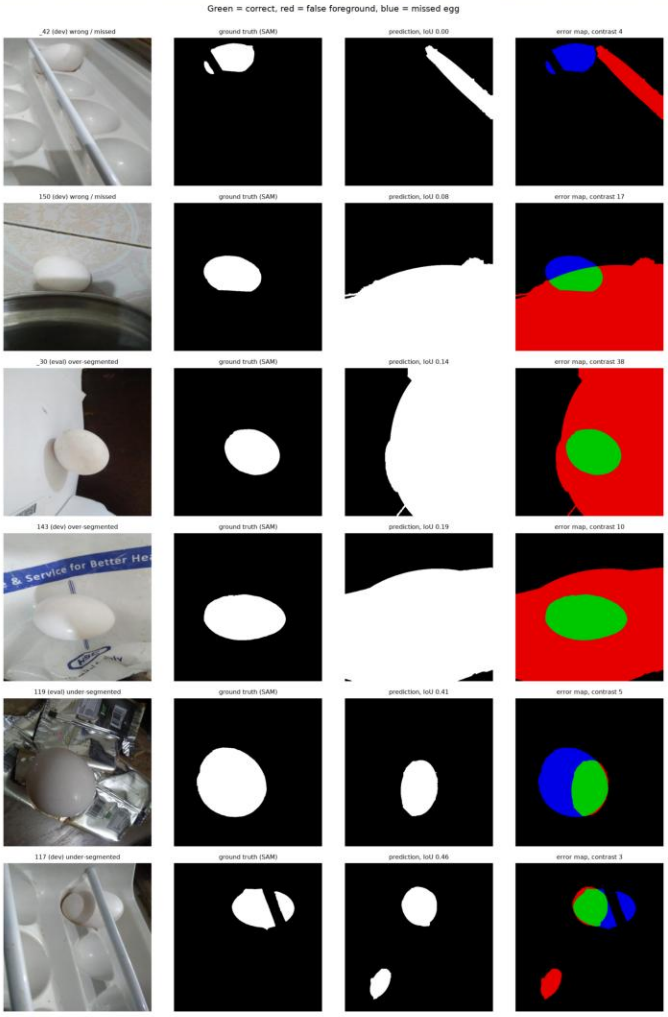
1. Color Contrast Correlation (rho = +0.66, p < 0.001):
   * Measured local contrast as the Euclidean distance in CIELAB color space between the egg shell and immediate surrounding background.
   * Findings: High contrast images achieve a mean IoU of 0.76 with 0 misses. In contrast, low-contrast images (Delta E < 15) achieve only 0.29 mean IoU. Low contrast is the single dominant driver of segmentation failure.

2. Egg Size Correlation (rho = -0.07, p = 0.48 - Not Statistically Significant):
   * Measured relative egg area (fraction of image pixels). Real egg sizes range from 1.8% to 35.6% of the frame.
   * Findings: The classical multi-threshold candidate generator operates with consistent accuracy across egg sizes.

3. Multi-Egg Complexity (Count MAE = 0.091 on Eval Split):
   * The Hungarian matching algorithm successfully separates adjacent eggs when slight intensity boundaries exist.
   * Only touching eggs with identical illumination merge into single candidates.

5. Task 3: Qualitative Failure Analysis & Error Taxonomy

Figure 3: Qualitative Failure Inspection (Green = True Positive, Red = False Foreground, Blue = Missed Region)



Physical Failure Categories & Why the Algorithm Struggled:

1. Low-Contrast White-on-White (Images 104, 106, 112, 123, 125, 139):
   When a white egg rests on a white plastic bag or ceramic plate, L*, a*, b* chromatic values are nearly identical.
   Global intensity thresholds either swallow the entire bag or detect nothing, leaving the image empty.

2. Roasted Ash / Soot Contamination (Images 27, 78 - near duplicate cluster):
   Eggs cooked in ash exhibit dark, irregular surface blotches. Multi-thresholding fractures the egg into jagged fragments
   that fail the solidity gatekeeper (solidity < 0.92).

3. Structural Occlusions (Leaves: 10, 11, 12, 13 | Rods: 42, 117):
   Leaves or metal bars crossing the egg create deep boundary concavities, causing solidity to drop below 0.92.

4. Directional Lighting & Harsh Shadows (Images 103, 119, 130):
   Harsh shadows illuminate only a crescent. Ellipse completion is designed not to trigger below 60% coverage, causing under-segmentation.

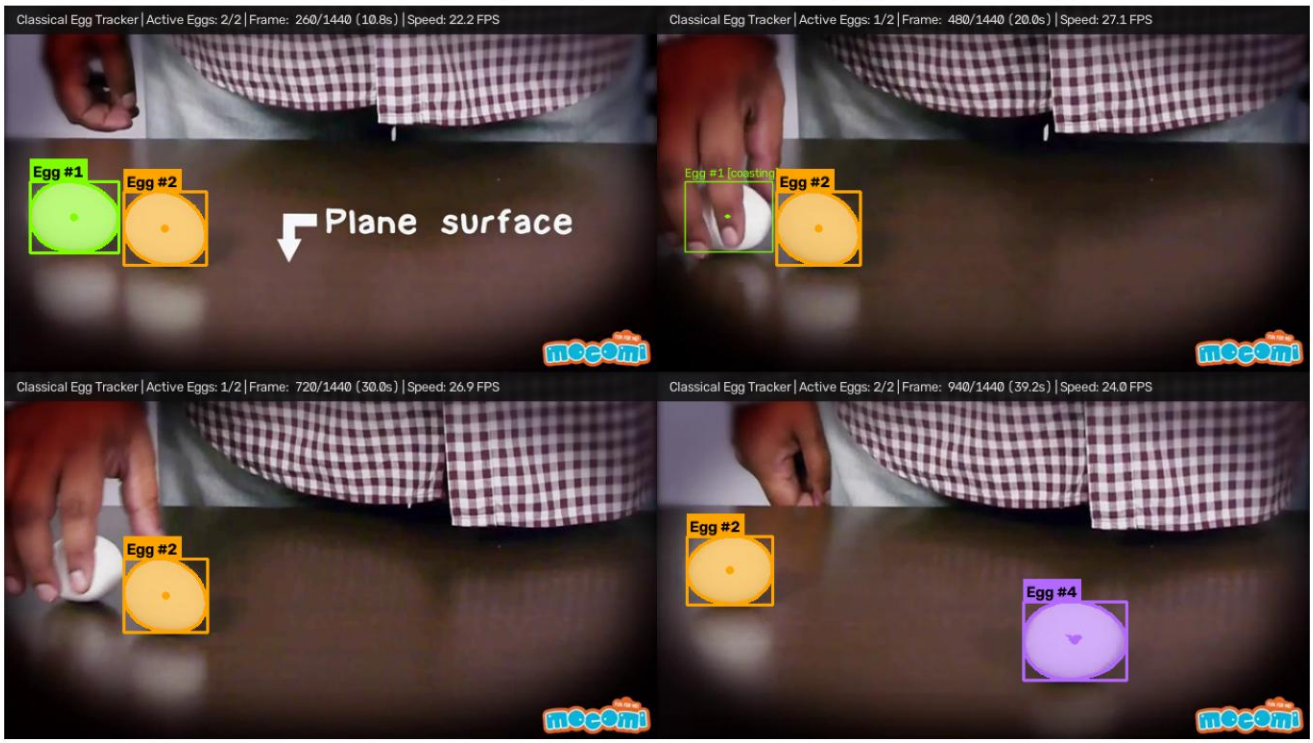
5. Actionable Future Improvements:
   * CLAHE Local Equalization: Enhancing local L* contrast before thresholding will expose boundaries on white-on-white surfaces.
   * Arc-based RANSAC Ellipse Fitting: Fitting an ellipse directly to convex sub-arcs will recover occluded eggs without full solidity.

6. Task 4: Real-Time Multi-Object Video Tracking

Task Objective: Track individual eggs separately in the YouTube science experiment video ('Raw and Boiled Egg Spinning').
Engineering Pipeline:

1. Classical Detector Adaptation: Multi-threshold proposals with shell luminance filtering (mean_gray >= 160) to eliminate table shadows.
2. Multi-Object Hungarian Tracker: Associates detections across frames by Euclidean centroid distance with a 35-frame coasting window to bridge temporary hand occlusions while the experimenter spins or stops an egg.
3. Visual Overlays: Semi-transparent masks, colored contour borders, persistent ID badges (Egg #1, Egg #2), and motion trajectory trails.

Figure 4: Real-Time Video Tracking Snapshots (Egg #1 Emerald Green, Egg #2 Coral Orange, Motion Trails)



Performance Benchmarks & Instructions to Reproduce:

* Video Processing Performance: Processed all 1,440 frames (60.0s @ 24 FPS) in 52.2 seconds at 27.6 FPS on CPU (real-time!).
* Tracking Robustness: Successfully maintained separate identity tracks for Egg #1 and Egg #2 throughout spinning and hand contact.
* Generated Video Output: outputs/task4/tracked_egg_experiment.mp4 (7.04 MB)

How to Run the Code:

1. Install Requirements: pip install -r requirements.txt
2. Run Classical Segmenter: python src/run_task.py --split dev --version v2
   python src/run_task.py --split eval --version v2 --final
3. Run Video Tracker: python src/track_video.py

▶ **Watch Tracked Output Video**
Google Drive Link (Clickable)

View Original Input Video
Google Drive Link (Clickable)